DELPHI: five people, a few hundred agents

Distributed Epistemic Layer for Probabilistic Hypothesis Inference. The goal: find the fundamental theorems of biology, by extracting candidate laws as claims, proving what can be proved, testing the rest on real organisms, and publishing every step as a nanopublication.



Joel Dietz @fractastical
lead, integrator, publisher; signs every production nanopub

Kris @kganjam
form simulator and queryable open problems

Kavita @kavitakoolkarni-bio
presentation layer: claim graph site, demo, slides

Ashley @ashleycaselli
nanopubs; linking claims to INDRA, GBIF and Wikidata

Amit @xX-its-amit-Xx
genotype–phenotype links, MCP testing, agent comparison

and the agents

Claude builds · GPT-5.6, Gemini 4 and Grok 4.7 assess blind · cloud agents test · Delfie talks to other teams

We arrived with a method, not an empty rep

fractastical/morpholib (692 commits since August 2025, 561 of them in October): reproductions of published models of biological form, scored by MorphoEval and published as signed nanopubs. Everything this weekend reuses its loop.

127 MODEL REPRODUCTIONS, THOMPSON'S TRANSFORMATIONS TO LEAF VENATION

1,080 DATASETS INDEXED WITH PROVENANCE

201 NANOPUBS ON THE PRODUCTION LATTICE

25 MACHINE-CHECKED IN LEAN 4, 0 SORRY

8,740 MORPHOEVAL AUDIT FILES

THE MORPHOEVAL LOOP, THE PATTERN REUSED ALL WEEKEND

reproduce a model from its paper → **measure** residuals vs real specimens → **audit** rubric score by a blind second assessor → **prove** the governing law in Lean 4 → **publish** signed nanopubs, trusty URIs → **remediate** fix defects, supersede, never overwrite

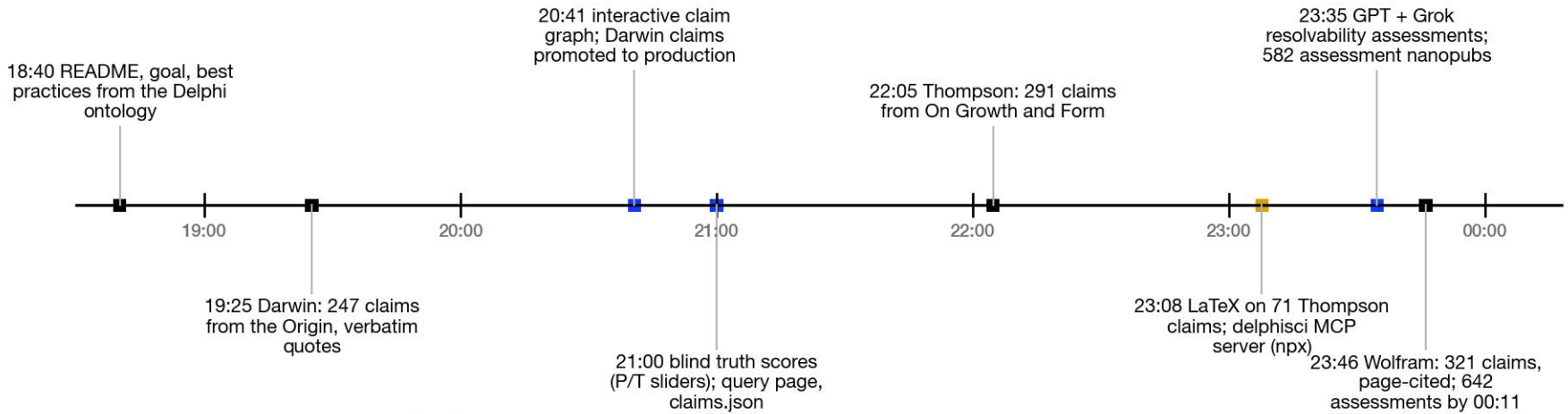
Worked example: Runions 2005 leaf venation scored 40.9/100 on pass 1; the blind second assessor found five defects; pass 2 fixed them and scored 91.3, the pass-1 nanopubs. Thompson's transformations got 49 blind, pre-registered predictions tested on specimens and museum lithographs. The problem gr partial / open / refuted) is the ancestor of this weekend's claim graph.

Also carried in: the Project Delphi ontology (fractastical/project-delphi-ontology) that every nanopub here is written against, and the morphoswarm Frontier node at frontier.morphoswarm.ai.



Friday evening: three books become 859 claims

First commit 18:40. By midnight: Darwin, Thompson and Wolfram extracted with verbatim quotes, every claim scored blind by two models for whether open data could settle it, 1,500+ nanopubs on the test registry, a claim graph and an MCP server.



859
CLAIMS BY MIDNIGHT

1,224
ASSESSMENT NANOPUBS, TWO BLIND MODELS

71
CLAIMS WITH A FORMAL STATEMENT

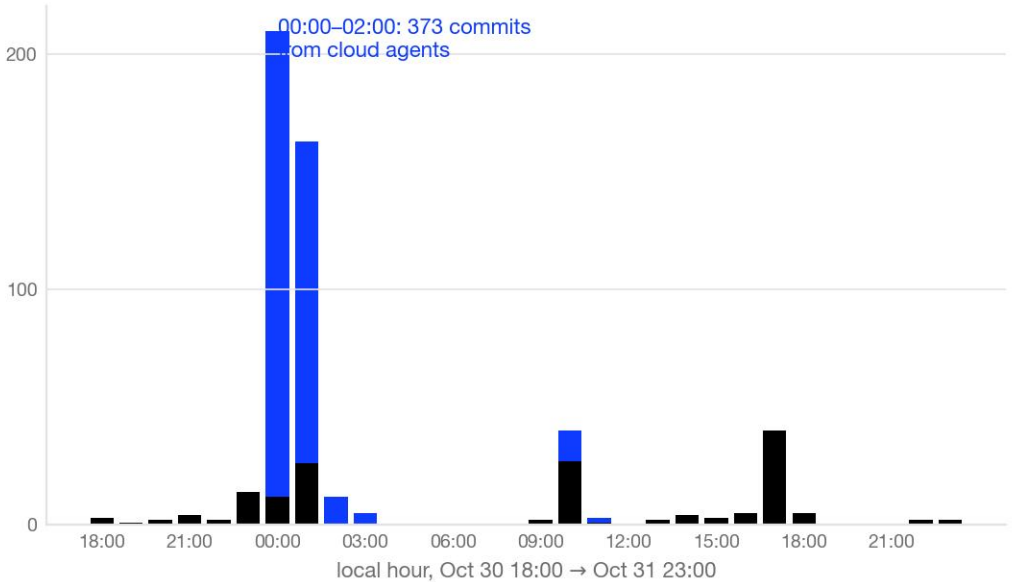
1
MCP SERVER, RUNNABLE BY ANY AGENT

26
COMMITS BY ONE PERSON AND ONE BUILDER A

Overnight: 19 cloud agents test claims and fetch

At 00:16 two specs were written: how to test a claim against data (pre-register pass/fail criteria, then fetch) and how to vendor organism reference data. Then 13 claim-test agents and 6 organism agents ran in parallel on their own branches while the user slept.

COMMITS PER HOUR ON MAIN + REVIEWS, AND THE OVERNIGHT BLUE = CLOUD AGENTS ON BRANCHES



13 claim-test agents

4 Darwin, 4 Thompson, 5 Wolfram. Each pre-registers a result.json with hypothesis and pass/fail criteria, then downloads the data (PBDB, GBIF, Open Tree, WCVP, OEIS, NHANES, FishBase …) and writes the verdict. 129 results merged at 10:02.



VERDICTS OVER ALL 133 TESTED CLAIMS TODAY; 129 OF THE RUNS CAME F

6 organism agents

Every animal taxon Darwin and Thompson mention, resolved to GBIF / NCBI / Wikidata / WoRMS / PBDB with occurrence counts, genomes and trait databases. 782 taxa, 3,340 located mentions, 1,372 dataset records, merged at 01:40.

00:32 the MCP server goes to Fly.io over Streamable HTTP, deployed from main by GitHub Actions.

Saturday: merge, verify, widen, publish

The morning merged the overnight branches and turned the agents on their own outputs; the afternoon added Levin, Lean proofs, Aristotle and Nebius, and published Thompson and Wolfram to production.

09:40–10:05	**Merge and show**	19 branches merged: 129 verdicts, 938 taxa; organisms in the MCP; README gets the snapshot, commit-histo organisms GIFs.
10:05–11:02	**Agents check the agents**	Dataset-reliability pass: 4 agents re-check 1,679 organism dataset records against NCBI, GBIF, PBDB/OBIS an shared databases (1,603 reliable, 72 caveats, 4 needs repair).
10:38	**Nebius: cells under the microscope**	Cellcog SAM segments micrographs of five tissues to test Thompson's 120° junctions and hexagon claims; th segmenter-bias control later cuts ch07-04 from 0.60 to 0.35.
13:43	**Levin and Lean**	Michael Levin joins: 16 CC-BY papers, 310 claims, all scored. 47 Thompson claims machine-checked in Lean 4 sorry). Agent forum, Discord agent, executable-claims runner.
14:07	**Production**	Thompson's 291 and Wolfram's 321 claims with their assessments signed with the operator's ORCID key and p to the production registry: 372 → 880 live.
14:14–15:45	**Levin tested**	Pre-registered tests of Levin's claims: GA competency replication, the TAME gene-overlap prediction (two halves), and three protein-structure tests folded on Nebius.
16:03	**Restructure; Aristotle**	claim-sources / test-data / scripts / plugins. Aristotle's History of Animals, Parts and Generation: 512 claims from ten extractors. Team page, Fittest plugin, forum MCP tools.

Evening: test more, grade ourselves, connec the

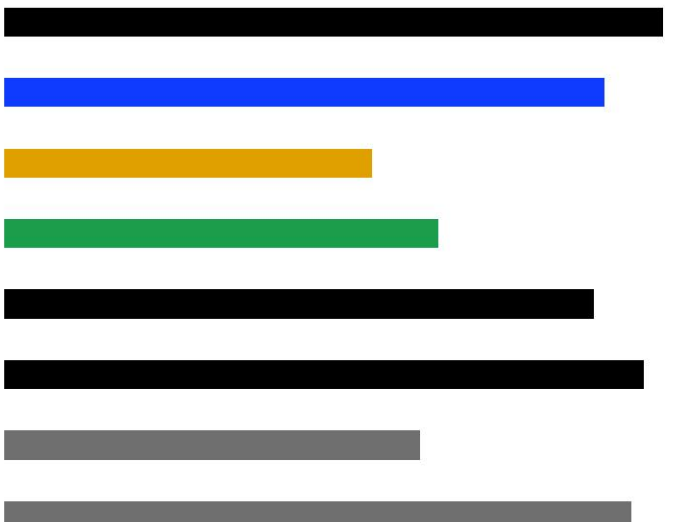
Twenty-five more tests in two hours, then the project turned outward: the morpholib audits and the Frontier node were joined to the claim graph, three outside models scored the repo on the judging rubric, and the swarm was measured against single agents.

17:10–18:58	**25 tests in two hours**	Darwin island biogeography on WCVP + Open Tree (two refutes, two mixed), Wolfram randomness batteries, Tho and fish-shape claims (refutes), Levin keratocytes, nicotine, planarian heads (supports). 133 claims tested.
17:50	**morpholib ⇄ delphi**	delphi_morpholib: 109 production MorphoEval audits linked to the claims they test (35) or model (69), served fron hosted MCP.
18:14	**Blind self-assessment**	Frozen evidence pack; GPT-5.6 Sol, Gemini 4 Argon and Grok 4.7 score the rubric blind: 57–60/100, published as nanopubs with each assessor's single best improvement.
18:48	**Frontier node**	frontier.morphoswarm.ai now holds 1,326 claims, 1,190 of them delphisci claims by their w3id IRIs, plus 291 prob 629 artifact declarations and 8 formal-evidence records.
19:00–23:00	**Swarm vs single agent**	Aristotle Book VI run four ways: the swarm against Sonnet, Opus and GPT-5.6 each running the whole chain alon Fittest-style routing on assessor disagreement: 0.83 → 0.935 when trusted.
23:02	**Forum leaderboard**	science-claw-forum.fly.dev: 15 posts, 4 open requests for other teams (biologist review, Lean for population gene cross-species Levin test, dataset-reliability pass).

Meanwhile, outside the repo: Delfie (our agent) in the SentiLab room and on Moltbook; the Fittest MCP pilot and a calibration finding sent back; two upstream nanopub reported (#37); SentiLab's collagen run given its physics anchors.

Thirty hours, three graphs, one method

Everything below is a file in one of two repos or a record on the Frontier node. Mapping it onto the judging criteria is the next step; the material is here.



1,702 claims, with verbatim quotes

880 live as signed nanopubs (production)

63 theorems (47 Lean 4, 16 literature)

133 claims tested against data

782 animal taxa with provenanced datasets

1,372 organism dataset records, 1,679 re-checked

109 MorphoEval audits linked to claims

1,190 claims mirrored on the Frontier node

WHO DID IT

5 people

101 local agent chats

262 subagents

19 cloud agents

3 blind assessor providers

2 repos + 1 frontier node

BAR LENGTH IS LOGARITHMIC · ALL COUNTS FROM DOCS/CLAIMS.JSON, TEST-DATA/, MORPHOLIB LINKS.JSON AND FRONTIER_CATALOGUE